



BRNO UNIVERSITY OF TECHNOLOGY

VYSOKÉ UČENÍ TECHNICKÉ V BRNĚ

FACULTY OF INFORMATION TECHNOLOGY

FAKULTA INFORMAČNÍCH TECHNOLOGIÍ

DEPARTMENT OF COMPUTER GRAPHICS AND MULTIMEDIA

ÚSTAV POČÍTAČOVÉ GRAFIKY A MULTIMÉDIÍ

COMPUTE-AWARE MOE FOR EFFICIENT 5G NEURAL RECEIVERS

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KNN PROJECT

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AUTHOR

AUTOR PRÁCE

**DOMINIK HUML (XHUMLD00)
JAKUB KONTRIK (XKONTR02)
MARTIN VACULIK (XVACULM00)**

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Chapter 1

Compute-Aware MoE for 5G Neural Receivers

1.1 Introduction

A 5G physical-layer receiver maps a received OFDM resource grid to soft bit estimates for the transmitted bits. In this project, BLER is measured directly on the recovered grid bits. A slot is erroneous if any of the predicted bits differs from the generated label. Recent work has shown that fully-neural receivers can match or surpass classical pipelines, particularly under hardware impairments and fast-varying channels [4, 9]. Their drawback is uniform compute. Every slot pays the same cost regardless of channel quality, although many slots are comparatively easy while a smaller set of difficult slots dominates the residual error rate.

Mixture-of-Experts (MoE) architectures [7, 3] provide a mechanism for per-sample adaptive compute. A router selects a subset of experts to run for each input. Prior work on MoE for 5G receivers, MEAN [2], showed hard-gated adaptation for SIMO neural receivers. Related wireless MoE work has used channel conditions in the gating decision [8]. Together, these results support expert routing as a natural mechanism for channel-dependent computation.

Our contribution is a compute-aware neural receiver that combines (i) heterogeneous experts, (ii) a channel-aware router driven by shared-stem features with no oracle SNR access, and (iii) an explicit FLOPs penalty in the training loss. The primary metric is the *BLER vs Average FLOPs* Pareto frontier on a held-out test set. We train and evaluate on synthetic 3GPP UMa and TDL-C channels generated with NVIDIA Sionna [5], using 128 subcarriers, 14 OFDM symbols, 16-QAM, and four receive antennas.

1.2 Related Work

Neural receivers for 5G

DeepRx [4] and NRX [9] establish that fully-neural physical-layer receivers can match or surpass classical pipelines, particularly under hardware impairments and fast-varying channels. Both apply a single dense network uniformly to every slot, which motivates our compute-aware comparison.

Mixture-of-Experts for wireless

The most relevant prior work for our setting is MEAN [2], a hard-gated MoE for a SIMO neural receiver. It demonstrates runtime adaptation and reports reduced active layers. Our work focuses on a different axis: we train heterogeneous experts (nano, small, large at 320, 695, and 1 604 M FLOPs) with a stem-based router and an explicit FLOPs penalty, which lets us trace a BLER-FLOPs Pareto frontier (Section 1.5). Song et al. [8] study channel-aware gating for distributed wireless MoE, where link conditions influence expert selection. We use the same broad principle with receiver-side features inferred by the shared stem.

Mixture-of-Experts training mechanics

The sparse expert formulation is from Shazeer et al. [7], with the Switch Transformer [3] contributing hard top-1 routing and a load-balance auxiliary. Differentiable routing uses the Gumbel-Softmax relaxation [6]. Section 1.5 tests these common anti-collapse mechanisms.

Simulators and data

Synthetic 3GPP UMa and TDL-C training and test data are generated with NVIDIA Sionna [5]. The DeepMIMO [1] ASU Campus 1 ray-traced scenario provides the out-of-distribution test set used in Section 1.7.

1.3 Architecture

The model has three components. A shared stem extracts channel-quality features, a channel-aware router picks one expert per sample, and three heterogeneous expert heads produce bit log-likelihood estimates. Figure 1.1 shows the layout.

Each slot is a 128×14 resource grid encoded as 16 input channels from received samples and LS channel estimates, using real and imaginary parts over four antennas. The model appends two pilot-distance maps and outputs $4 \times 128 \times 14 = 7168$ bit LLRs. BLER is computed after zero-thresholding these logits, with one block error if any bit is wrong.

The input follows a single-stream SIMO model. For receive antenna r , subcarrier k , and OFDM symbol t , the received resource element is generated as $y_{r,k,t} = h_{r,k,t}x_{k,t} + n_{r,k,t}$. The transmitted data symbol $x_{k,t}$ is unknown on data positions, but a sparse set of pilot positions is fixed and therefore known to the receiver. In our generator these pilots use the 16-QAM symbol corresponding to all-zero bits, placed every second subcarrier on OFDM symbols 2 and 11. The LS estimate divides the received pilot values by the known pilot symbol, then linearly interpolates the complex channel over frequency and time. Thus the neural receiver is given both the noisy received grid and a conventional pilot-derived channel estimate, not an oracle channel.

Shared stem

A three-layer 2D CNN with channels $18 \rightarrow 64 \rightarrow 64 \rightarrow 56$ and 3×3 convolutions maps the per-resource-element tensor to `state_dim=56`. The first 16 channels are the real and imaginary parts of y and of the LS estimate over four receive antennas. The remaining two channels are deterministic pilot-distance maps, one in frequency and one in time, which tell the network how far each resource element is from the nearest pilot location. The stem costs 285 M multiply-add FLOPs per slot.

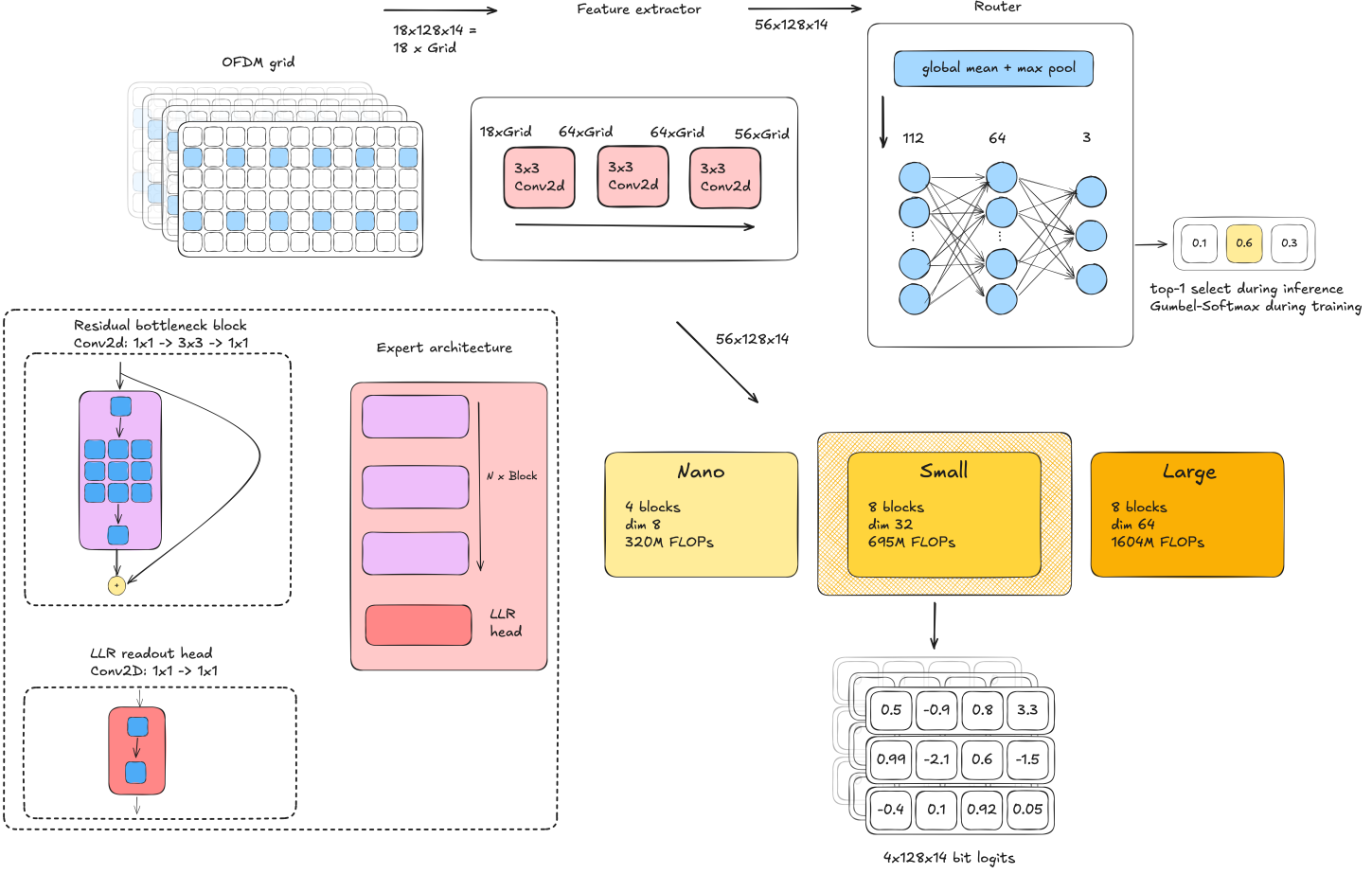


Figure 1.1: Compute-aware MoE neural receiver. The shared stem is always paid. The router selects exactly one expert per sample at inference. Experts have heterogeneous capacities (1604 M, 695 M, and 320 M total FLOPs).

Channel-aware router

A two-layer MLP ($112 \rightarrow 64 \rightarrow 3$, ReLU) maps mean-pooled and max-pooled stem features to expert logits. During training, the router is evaluated through a differentiable Gumbel-Softmax relaxation [6]. At inference the router selects exactly one expert with hard top-1 and has no access to the true SNR.

Heterogeneous experts

Each expert is a stack of residual bottleneck blocks ($1 \times 1 \rightarrow 3 \times 3 \rightarrow 1 \times 1$), followed by a bit-LLR readout head and a channel-estimate readout head. The bit head maps the expert state to four logits per resource element, one for each 16-QAM bit. The separate channel head is only an auxiliary training head. It maps the same state to 2×4 real-valued channels, representing the real and imaginary parts of the four-antenna channel. The three configurations are listed in Table 1.1.

The complete model has 0.583 million trainable parameters. Dense receivers at the corresponding capacity points (0.090, 0.168, and 0.450 million parameters) serve as base-

Expert	Blocks	Block dim	Readout dim	Total FLOPs	% of large
nano	4	8	32	320 M	20 %
small	8	32	96	695 M	43 %
large	8	64	128	1604 M	100 %

Table 1.1: Heterogeneous expert configurations. *Total FLOPs* is the sum of the always-paid stem (285 M), the expert backbone, and the two readout heads. Values verified analytically and against the model’s logged `max_flops` buffer to within 0.01 %.

lines. The final recipe uses them to warm-start the stem and the `nano`/`small` experts in Section 1.5.

Training objective

The loss combines the bit-level binary cross entropy with a channel-estimate auxiliary, an expected-FLOPs penalty, and a soft load-balance term,

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \gamma \mathcal{L}_{\text{chan}} + \alpha \mathbb{E}_{\text{router}}[\hat{C}] + \beta \mathcal{L}_{\text{bal}},$$

where $\hat{C}$ is the total per-slot cost, stem included, normalised by the always-**large** receiver. The channel term is an MSE between the channel head output and the simulator’s true complex channel, stored as real and imaginary parts. This target is available only because the training data are synthetic. It is not part of the deployment input and is not used by the BLER metric. In the objective it acts as a supervised auxiliary term on the same expert state from which the bit head predicts LLRs. We use $\gamma = 0.05$, $\alpha = 2 \cdot 10^{-3}$ at the final operating point, and $\beta = 0.1$. The expected-FLOPs term uses soft routing during training, while realised FLOPs use hard top-1 routing at inference.

1.4 Data and Evaluation Protocol

Synthetic training data

The final alpha-sweep recipe uses a 12 000-step budget on the mixed UMa+TDL-C profile. The training loader preloads a local Arrow subset and alternates batches between the two profiles. For the selected operating point, the submission caps the subset at 50 000 samples per profile. With batch size 128, each full run processes up to 1 536 000 sample presentations, split evenly between UMa and TDL-C, over repeated reshuffled passes through the same 100 000 stored examples. Reported test numbers use the best checkpoint selected within this budget.

The synthetic profiles share the same receiver format. Each generated sample uses 16-QAM, four receive antennas, 128 subcarriers, 14 OFDM symbols, 15 kHz subcarrier spacing, a 3.5 GHz carrier, pilot-based LS channel estimation, and the same resource-grid tensor consumed by the neural receivers. UMa is generated as a 3GPP Urban Macro uplink with a 500 m cell radius, 25 m base-station height, 1.5 m user height, and user speeds up to 35 m/s. TDL-C uses the 3GPP tapped-delay-line C model with 500 ns delay spread and speeds up to 25 m/s. The locked test split contains 32 768 samples per profile and is used only after training.

DeepMIMO out-of-distribution data

DeepMIMO is not used for optimisation. The ASU Campus 1 split is a ray-traced channel dataset, so it tests a different channel distribution while keeping the receiver input format unchanged. Our generator selects DeepMIMO channel responses, extracts the same 128 subcarriers and four receive antennas, normalises channel power, then applies the same 16-QAM symbols, additive noise, pilots, LS interpolation, and bit labels as the Sionna generator. The resulting ASU test split contains 32 768 samples and is stored as an Arrow dataset. This makes the DeepMIMO experiment a domain-shift check, not a change in modulation, tensor shape, or evaluation metric.

Item	Protocol
Train	UMa+TDL-C, 50 000 stored samples per profile, 12 000-step budget, batch 128
Test	locked split, 32 768 samples per profile
OOD	DeepMIMO ASU Campus 1, 32 768 samples
Input	128×14 grid, four RX antennas, 16-QAM, LS pilots
Metric	BLER on hard bit decisions

Table 1.2: Training and evaluation protocol used for the final reported results.

1.5 Training Recipe

Here *cold* and *warm* refer only to parameter initialisation. A cold expert starts from random weights. A warm expert is initialised from a dense receiver checkpoint with matching dimensions. The implementation copies dense `stem.*` weights into the shared stem and copies dense `backbone.*`, `readout_llrs.*`, and `readout_channel.*` weights into the corresponding expert branch. The router itself remains newly initialised.

Two degenerate regimes appear when training the compute-aware MoE. A fully cold start lets the FLOPs penalty push routing toward cheap experts before **large** can recover blocks (BLER 0.926 at 48 % FLOPs). A full warm-start initialises the stem and all three experts from dense checkpoints, which gives **large** the initial advantage and yields a fine-tuned dense receiver (BLER 0.879 at 100 % FLOPs). Figure 1.2 contrasts these regimes with our asymmetric warm-start recipe.

Anti-collapse mechanisms

We tested four Switch auxiliary weights from 10^{-3} to 10^0 and four soft capacity penalties from 0.1 to 10.0. None of the eight runs produced both heterogeneous routing and BLER in the selected MoE range. The three smaller Switch weights collapsed to **large**. The largest weight spread soft probabilities, but hard top-1 routing still selected **large** for nearly all samples. A weak capacity penalty also collapsed, medium penalties spread routing but degraded average BLER by about 6–7 pp, and capacity weight 10.0 recovered only a worse operating point than the selected MoE. Two warmup variants gave the same conclusion. Large-warmup routed every tested seed to **large**, while β -warmup produced inconsistent routing and worse average BLER than the no-warmup baseline.

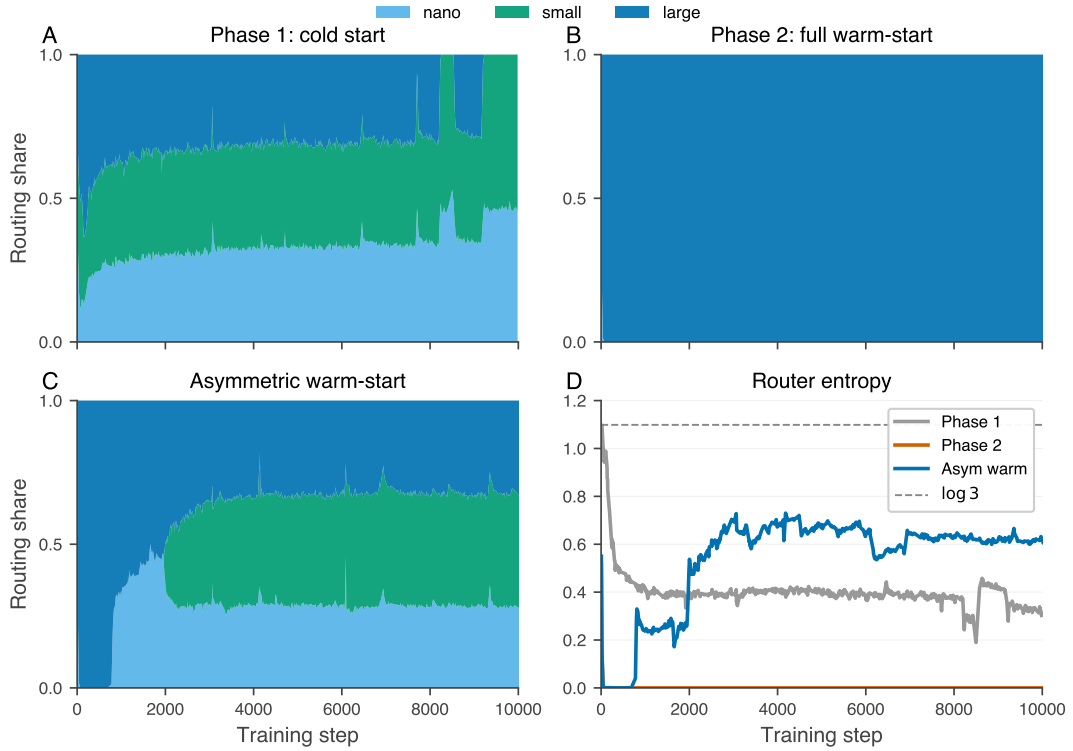


Figure 1.2: Routing trajectories for the three paradigms. Phase 1 abandons **large** early, while Phase 2 converges to 100 % **large**. The asymmetric warm-start recipe maintains exploratory router entropy long enough to reach a heterogeneous policy.

Asymmetric warm-start with cold-large.

Our recipe warm-starts the stem, **nano**, and **small** from dense baselines while leaving **large** random. This is deliberately asymmetric. It keeps the small experts immediately useful, but removes the early competence advantage that otherwise makes the router send nearly all samples to **large**. In Figure 1.3, cold **large** initially captures routing, **nano** catches up around step $\sim 1\,500$, **small** joins around $\sim 1\,700$, and from $\sim 3\,000$ onward all three experts remain active.

Why cold-large specifically.

The cold-**small** variant produces heterogeneous routing but at higher cost (validation BLER 0.897, FLOPs 67 %). The cold-**nano** variant routes essentially all samples to **large** (validation BLER 0.887, FLOPs 100 %). In this sweep, cold-**large** is the configuration that makes the router commit to smaller experts before **large** becomes competent.

The α sweep on the test set

Table 1.3 reports test-set BLER and average realised FLOPs for the four operating points (each 32 768 samples per profile). The knee of the Pareto curve sits at $\alpha = 2 \cdot 10^{-3}$.

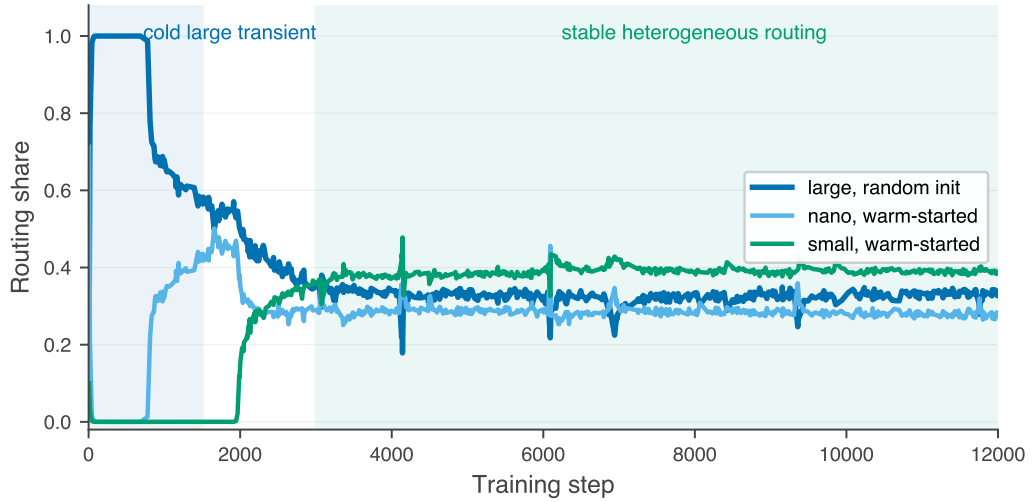


Figure 1.3: Expert routing share (EMA) for the selected MoE. Cold-initialised **large** dominates briefly at the start, then drops as the FLOPs penalty pushes the router toward the warm-started **nano** and **small**. From step $\sim 3\,000$ onward the routing is stably heterogeneous across all three experts.

Operating point	α	UMa BLER	TDLC BLER	Avg FLOPs	TDLC routing (l/n/s)
Dense large baseline	n/a	0.936	0.866	100 %	n/a
$5 \cdot 10^{-4}$ setting	$5 \cdot 10^{-4}$	0.936	0.861	100 %	100/0/0
$1 \cdot 10^{-3}$ setting	$1 \cdot 10^{-3}$	0.938	0.875	56 %	44/12/44
selected MoE	$2 \cdot 10^{-3}$	0.937	0.867	56 %	44/15/40
$5 \cdot 10^{-3}$ setting	$5 \cdot 10^{-3}$	0.940	0.881	60 %	37/0/63

Table 1.3: Alpha sweep on the locked test set. Average realised FLOPs is taken over UMa+TDLC, normalised to the always-**large** cost (1 604 M). The lowest- α setting routes entirely to **large**. The highest- α setting assigns no test samples to **nano**.

Channel-aware feature ablation

Replacing the router input with fresh Gaussian noise raises test BLER to 0.968 (+6.6 pp) and sends 89 % of samples to **small** with 0 % **large** usage. This supports channel-quality input as the active mechanism, rather than the architecture or loss alone.

Linear probes indicate profile-specific channel features

Linear probes on pooled stem features indicate profile-specific channel information. On TDL-C, SNR reaches $R^2 = 0.93$ and delay spread $R^2 = 0.65$. On UMa, the SNR probe is weaker ($R^2 = 0.42$), consistent with geometry-sensitive decodability. UMa-vs-TDL-C classification reaches 77.9 % accuracy.

Recipe stability is bimodal

Re-running the selected recipe with seeds $\{32, 42\}$ gave one reproduction (s42, validation BLER 0.905) and one degenerate run (s32, test BLER 0.958, **large** abandoned). At 100k samples, $\alpha = 2 \cdot 10^{-3}$ routes away from **large** for both s67 and s42. Scaling α down to

$1 \cdot 10^{-3}$ at 100k recovers the selected MoE quality, with test BLER 0.9013 at 58 % FLOPs. Recipe stability at fixed α remains open.

1.6 Training-Scaffold Finding

The selected MoE has three experts active in routing, but block-level success is concentrated entirely in the largest expert. This observation yields a deployment recipe that reduces FLOPs without retraining.

Per-expert success rate

For each routed sample we record whether the assigned expert recovered the whole bit grid without error. Across about 8000 test samples, **nano** and **small** both succeed on 0 % of routed samples. Only **large** succeeds, on 23.2 % of UMa-routed samples ($n=1047$) and 29.4 % of TDL-C-routed samples ($n=1832$). Figure 1.4 shows that routing is heterogeneous, but block success is not.

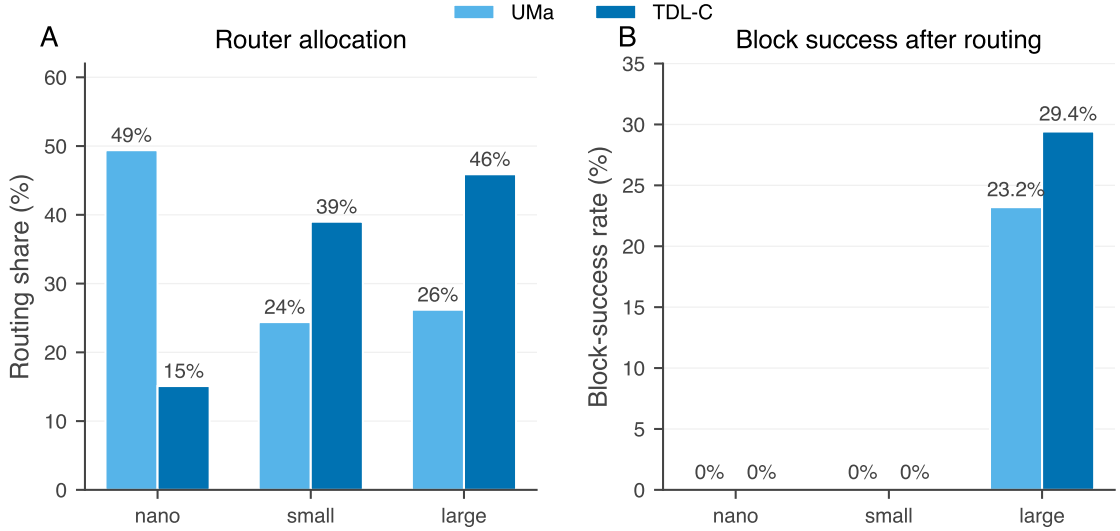


Figure 1.4: The training-scaffold finding. The left panel shows that the router uses all three experts. The right panel shows that only **large** recovers complete routed samples.

Counterfactual bit recovery on small-routed samples

For each test sample we forced every expert to recover the bit grid. On the UMa samples routed to **small** ($n=979$), hard-decision BER is nearly identical: **nano** 0.237, **small** 0.230, and **large** 0.231. BLER is 1.000 for all three. The router therefore sends this subset to **small** in a regime where no evaluated expert changes the block outcome. The routing decision reduces compute on samples that were not recovered by any evaluated expert, rather than recovering them through **small**.

Mode B substitutes `small` with a zero-output sink at inference.

Since `small` contributes no complete block recoveries in the routed evaluation, its expert computation can be removed. We replace its expert head with a constant zero output and charge zero FLOPs whenever the router selects `small`. No retraining is performed.

Operating point	Avg BLER		Δ	Avg FLOPs	
	base	Mode B		base	Mode B
$\alpha = 10^{-3}$	0.9063	0.9066	+0.0003	56.0 %	46.7 %
selected MoE	0.9021	0.9021	0.0000	55.8 %	47.3 %
$\alpha = 5 \cdot 10^{-3}$	0.9107	0.9116	+0.0009	60.0 %	41.9 %

Table 1.4: Mode B inference. On the selected operating point, BLER is unchanged to four decimals, and FLOPs drops by 8.5 pp. Across all three α values BLER drift stays at or below 10^{-3} .

Figure 1.5 places these operating points against the dense neural ladder and the LS-MRC reference.

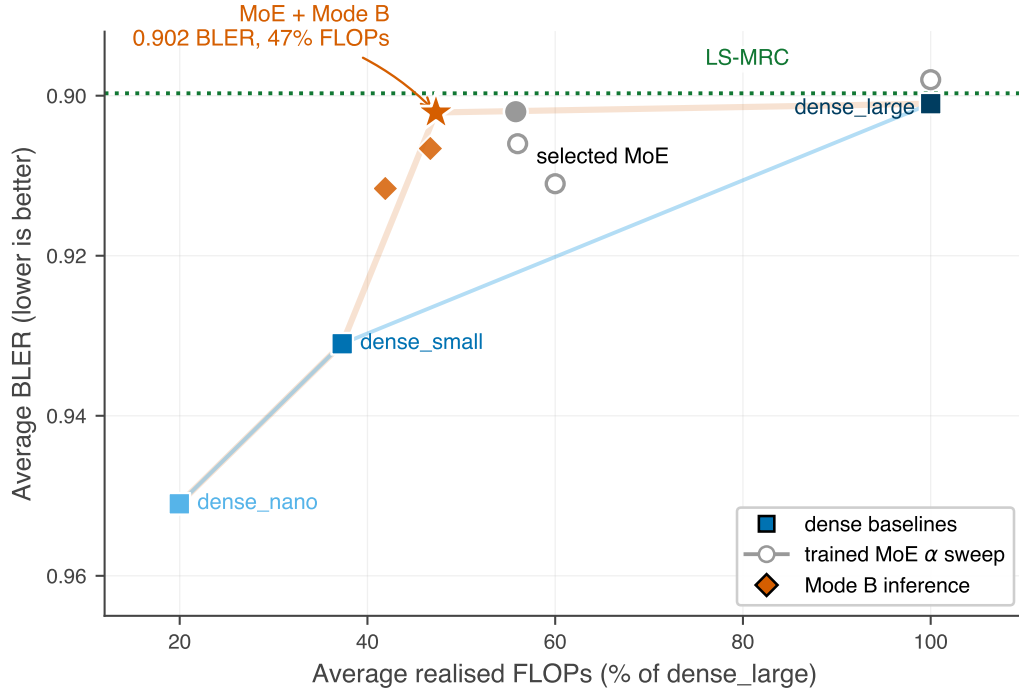


Figure 1.5: Pareto frontier on the locked test set. Dense neural baselines trade BLER for FLOPs. The trained MoE lands near `dense_large` at 56 % FLOPs, while Mode B shifts it leftward. **MoE + Mode B is within 0.002 BLER of `dense_large` at 47.3 % FLOPs.** LS-MRC is the classical reference and has no neural FLOPs.

Why `small` is needed during training.

The two-expert ablation drops `small` and trains `{nano, large}` under the same recipe. Validation BLER rises to 0.955, a +5.3 pp loss against the selected MoE under the matched

protocol. Removing **small** from training is therefore much more harmful than removing it at inference. The result indicates that **small** shapes the optimisation trajectory, even though its expert head is dispensable after training.

Role of **nano**.

The symmetric ablation drops **nano** and trains only **small** and **large**. It gives test BLER 0.909 at 65 % FLOPs, costing +0.7 pp BLER and +9 pp FLOPs versus the selected MoE. Since **nano** absorbs many samples whose block outcome does not change under larger experts, dropping it costs more in compute than in BLER.

Training with a sink expert

We also trained models with a zero-output expert from the start. The sink, channel-only, large variant ends at 82 % FLOPs. The sink, small, large variant moves to the all-sink solution by step 500 and reaches BLER 1.0. The sink, nano, large variant is stable but ends 7 pp BLER worse than the selected MoE. The sink expert is therefore useful post hoc, but not as a trainable expert under the same recipe.

Deployment interpretation

The resulting MoE is best understood as one bit-recovery expert, **large**, plus cheaper routing outcomes for samples where extra capacity does not change the block result. **small** is useful during optimisation but removable at deployment. *MoE + Mode B* achieves BLER 0.9021 at 47.3 % of dense-large FLOPs with no retraining.

1.7 Scope and Discussion

The contribution is positioned against dense neural receivers. We add scope checks against classical baselines, oracle routing, and ray-traced channels.

Full baseline landscape

For a fair compute-vs-quality comparison we evaluated three classical baselines and three dense neural baselines of varying capacity on the same locked test set (32 768 samples per profile). The classical receivers have no trainable parameters. The single-antenna baseline uses the first receive antenna only and therefore discards spatial diversity. LS-MRC uses the same received samples and pilot-derived LS channel estimate as the neural models, combines the four receive antennas by maximum-ratio combining, and converts the equalised 16-QAM symbol to max-log bit logits using the sample SNR for the noise variance. Genie-MRC repeats the same detector with the true simulated channel instead of the LS estimate, so it is an oracle upper bound rather than a deployable receiver. Table 1.5 reports the full landscape.

MoE + Mode B reaches dense_large-level BLER at $\sim 47\%$ of dense_large FLOPs. This is below half of the full neural receiver while remaining much more accurate than dense_small. Against classical baselines it has much lower BLER than single-antenna detection and is essentially tied with LS-MRC on average BLER. The oracle Genie-MRC receiver has the lowest average BLER, so the comparison with classical receivers is channel-dependent.

Baseline	Channel info	Avg BLER	UMa	TDL-C	Avg FLOPs
Single-antenna	none	0.9950	0.9922	0.9978	n/a
Genie-MRC (oracle)	true H	0.8544	0.9084	0.8003	n/a
LS-MRC	LS pilots	0.8997	0.9388	0.8607	n/a
dense_nano	LS+net	0.951	0.961	0.941	320 M
dense_small	LS+net	0.931	0.951	0.911	599 M
dense_large	LS+net	0.901	0.936	0.866	1604 M
MoE + Mode B (ours)	LS+net	0.9021	0.937	0.867	759 M

Table 1.5: All baselines on the same locked test set. Single-antenna is a no-MIMO floor. Genie-MRC needs oracle channel knowledge and is not deployable. LS-MRC is the realistic classical baseline. MoE + Mode B is within 0.002 BLER of dense_large while using less than half of dense_large’s neural FLOPs.

The strong LS-MRC result should be interpreted in the context of the evaluated single-stream SIMO 1×4 setting. In this regime, maximum-ratio combining is a well-matched classical receiver. It exploits receive diversity, has no learned parameters, and faces no multi-stream spatial interference. The result therefore does not imply that classical receivers dominate learned receivers in general. Within this benchmark, the proposed MoE remains close to LS-MRC while matching the dense neural receiver at substantially lower neural FLOPs. More complex MIMO settings, hardware impairments, imperfect calibration, and multi-user interference are natural follow-up settings, but they are not evaluated here.

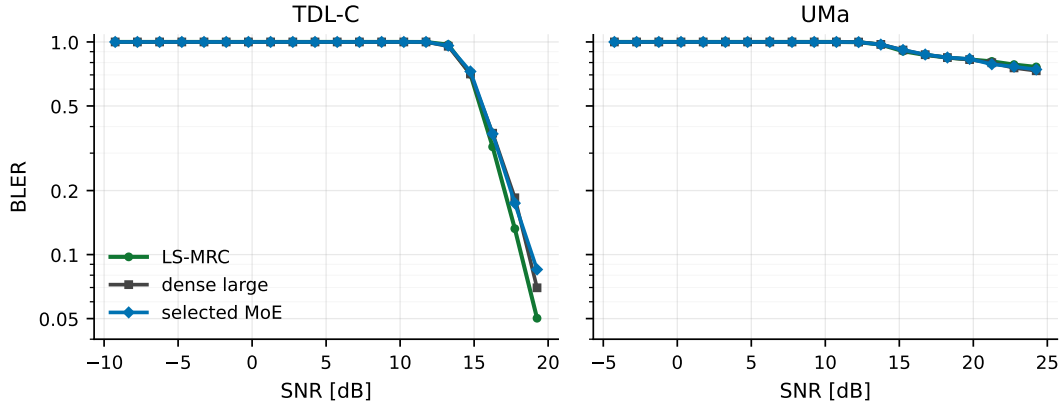


Figure 1.6: High-resolution SNR waterfall on the locked test split. All curves use 32 768 samples per profile. LS-MRC drops faster on the TDL-C transition region, while the selected MoE tracks dense_large closely. UMa remains difficult across the plotted SNR range, with the neural receivers only slightly below LS-MRC in the highest bins.

On three in-family OOD profiles (TDL-A, TDL-D, CDL-A), the selected MoE and dense_large both average 0.825, while LS-MRC averages 0.802. The MoE preserves dense_neural quality while reducing neural compute, but the classical receiver has lower BLER on these profiles.

An oracle-SNR cascade is a tighter Pareto reference

Beyond the dense ladder we compare against a hand-rule cascade using true SNR, an upper bound not deployable in practice. At tolerance $+0.01$ – 0.02 it reaches BLER 0.900 at 49 % FLOPs, slightly dominating learned routing (0.902 at 56 %). Looser tolerances reach 0.909 at 34 % and 0.919 at 27 %. The selected MoE remains on the Pareto frontier without oracle SNR access. A direct SNR-statistics router ablation collapsed to 100 % **large**.

Where neural receivers retain an advantage

UMa shows the opposite pattern from TDL-C. The receivers tie within 1 pp at low and mid SNR, while neural receivers have slightly lower BLER at $\text{SNR} \geq 20.5$ dB (Table 1.6). A UMa crosstab gives the selected MoE a net +64 block advantage over LS-MRC. On TDL-C, the same crosstab favours LS-MRC by 220 blocks. The UMa NN-favoured subset has weaker MIMO geometry, with mean spatial max-eigenvalue 2.07 against 3.22 and channel power 0.56 against 0.89. This is consistent with the learned receiver helping on samples where MRC aggregation is less favourable.

SNR (dB)	TDL-C waterfall			SNR (dB)	UMa high-SNR		
	LS-MRC	dense_large	MoE		LS-MRC	dense_large	MoE
14.0–15.5	0.7050	0.7069	0.7265	14.5–16.0	0.9047	0.9162	0.9168
15.5–17.0	0.3213	0.3722	0.3697	17.5–19.0	0.8443	0.8443	0.8449
17.0–18.5	0.1324	0.1850	0.1745	20.5–22.0	0.8100	0.7981	0.7856
18.5–20.0	0.0502	0.0698	0.0851	22.0–23.5	0.7825	0.7551	0.7651
				23.5–25.0	0.7629	0.7320	0.7423

Table 1.6: Per-SNR BLER, 32 768 samples per profile. LS-MRC has lower TDL-C BLER throughout the plotted waterfall region, while neural receivers have lower UMa BLER at $\text{SNR} \geq 20.5$ dB by 1–3 pp.

Out-of-distribution behaviour on ray-traced channels

On DeepMIMO ASU Campus 1 ray-traced channels the neural receivers have near-unit BLER (dense_large 0.9901, selected MoE 0.9915), so this split is a clear OOD failure case rather than a successful transfer result. The selected MoE also falls back mostly to **nano** at ~ 32 % FLOPs. In Figure 1.7, ASU samples form a tight off-centroid cluster in the in-distribution stem-feature PCA basis, with only 20 % inside the one-sigma box. A few-shot fine-tune (2048 samples, 500 steps) did not recover. As a separate DeepMIMO O1 control, LS-MRC is also poor (BLER 0.976), with dense_large at 0.982 and the selected MoE at 0.984. The DeepMIMO checks therefore expose a domain-shift limitation, not an additional claim of robust ray-traced generalisation. Longer fine-tuning, a larger OOD slice, and domain-randomised pre-training are left as natural next checks.

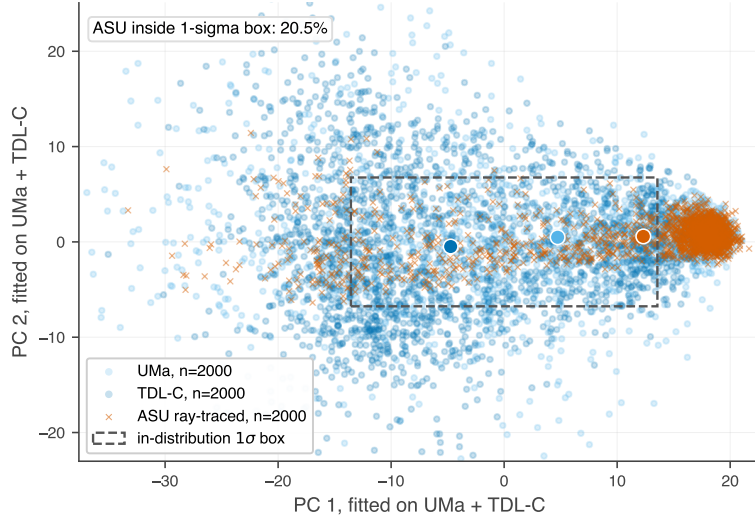


Figure 1.7: Stem-feature PCA-2D fitted on in-distribution UMa+TDL-C. ASU ray-traced samples (orange) projected onto the same basis collapse into a narrow off-centroid cluster (only 20 % inside the in-distribution one-sigma box), explaining the router’s **nano** default on OOD.

Conclusion

We presented a compute-aware MoE neural receiver. Per-expert success analysis showed that the middle expert did not recover any complete block in the evaluated setting. Replacing it with a zero-output sink at inference preserves BLER to four decimals while saving 8.5 pp FLOPs, with no retraining. Diverse experts are needed during optimisation, but the non-recovering middle expert can be pruned at deployment. Open problems include recipe stability and OOD transfer to ray-traced channels.

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